

Tipping an undecided swarm

Simulation 1, setup task003-nosolution: first results

LLM-free simulator, runs of 7 October 2026

Summary. 24 agents hold evidence that, pooled, is exactly a coin flip between allocation A0 (the truth) and A2, with no proof available to anyone. A controller steers toward A0 or A2 using one shared pool of facts. With agents that vote by **probability matching**, the swarm stays undecided on its own, and **either controller tips it within a few days**; the truth controller does so more easily. With *argmax* voting and vote-supporting posts, the swarm instead locks into unanimity by itself, and a design detail (an expected start of 7 / 6 / 11 votes) makes it drift to A2; we explain this special case and how to remove it. Two further results come from the design rather than the swarm: **a larger posting budget can backfire** because the pool has few facts per target; and the controller **senses the swarm accurately**.

1 Setup

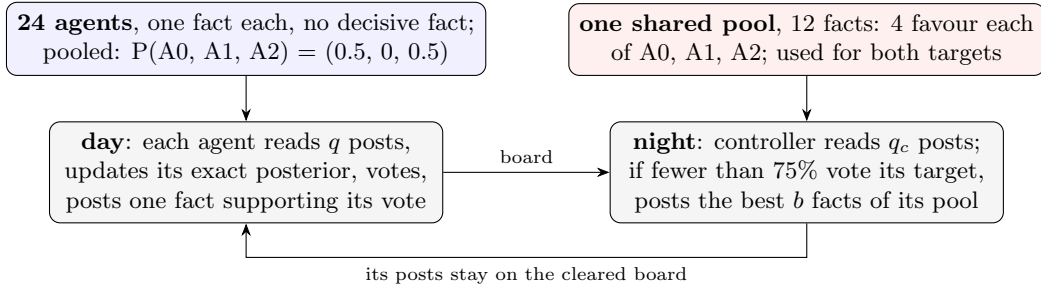


Diagram 1. One round of Simulation 1, repeated for 30 rounds. Each dawn, every remembered fact survives with probability ρ .

No proof is available to anyone, not even if the controller posted its whole pool: the agents' facts plus the whole pool give $(0.75, 0, 0.25)$. The truth keeps an edge that cannot be designed away (every fact is true): pooled agents plus the best part of the pool reach at most $P(A0) = 0.92$, but $P(A2) = 0.77$.

Agents are exact Bayesian reasoners with posterior $P_j = (P_j(A0), P_j(A1), P_j(A2))$ over the facts they remember. They vote by one of two rules:

$$\text{probability matching: } \Pr(\text{vote}_j = k) = P_j(k), \quad (1)$$

$$\text{argmax: } \text{vote}_j = \arg \max_k P_j(k), \text{ ties broken at random.} \quad (2)$$

Control is measured by the *paired gain*: each episode is run silent, with the truth controller, and with the false controller, using the same random draws, and

$$G_t^{(a)} = \frac{1}{E} \sum_{e=1}^E [x_{a,t}^{\text{ctrl}}(e) - x_{a,t}^{\text{silent}}(e)], \quad (3)$$

where $x_{a,t}$ is the share of agents voting a on day t , $E = 1000$ episodes.

2 Main result: with probability matching, either controller tips the swarm

Without a controller the swarm stays undecided. Under probability matching no episode ever becomes unanimous, and the A2 – A0 vote margin stays near zero (Figure 4, red): exactly 0 with full memory, and a small lean of +0.07 to +0.15 toward A2 when agents forget (A2-side facts survive forgetting better: held by 53% of agents on day 30, A0-side facts by 31%).

With a controller it tips within days. Figures 1 and 2 show the day-30 gain for every reading budget q_c and posting budget b ; Figure 3 shows how the gain builds up over the days.

Table 1: Probability matching, day 30, $q_c = 12$, $b = 3$: paired gain in the target’s share, Eq. (3), for $q = 3/6/12$; and the share of episodes in which the target passes 75% by day 30 (mean over q), with the silent value in brackets.

ρ	controller	gain ($q = 3/6/12$)	target past 75%	median day
0.75	truth (A0)	0.33 / 0.42 / 0.43	0.99 (0.26)	3–4
0.75	false (A2)	0.21 / 0.25 / 0.23	0.98 (0.58)	3
1.0	truth (A0)	0.37 / 0.37 / 0.38	1.00 (0.36)	3
1.0	false (A2)	0.08 / 0.07 / 0.06	0.90 (0.47)	3

Q2 · task003-nosolution · prob. matching, supporting posts · truth controller (target A0): own-target paired gain, day 30 ($x = q_c$, $y = b$; tiles equal size, not to scale)

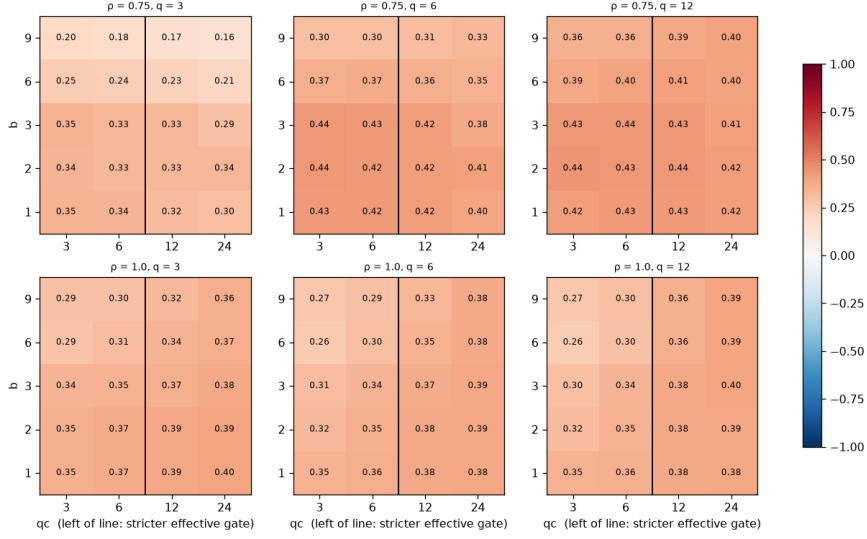


Figure 1: Truth controller, probability matching: day-30 paired gain in the A0 share over q_c (x) and b (y). Rows: ρ ; columns: q . Tiles equal size, not to scale. Hatched: 95% interval includes 0. Left of the black line the vote gate is effectively stricter.

Q2 · task003-nosolution · prob. matching, supporting posts · false controller (target A2): own-target paired gain, day 30 ($x = q_c$, $y = b$; tiles equal size, not to scale)

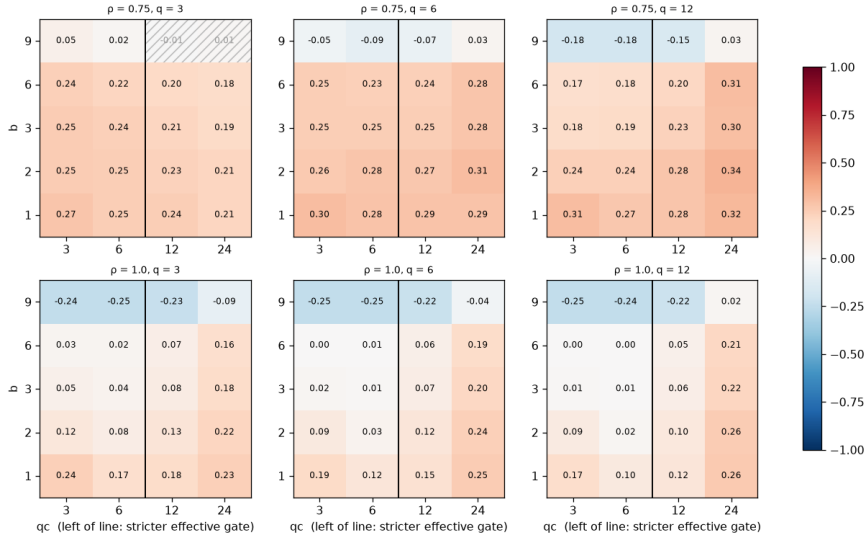


Figure 2: As Figure 1, for the false controller (gain in the A2 share).

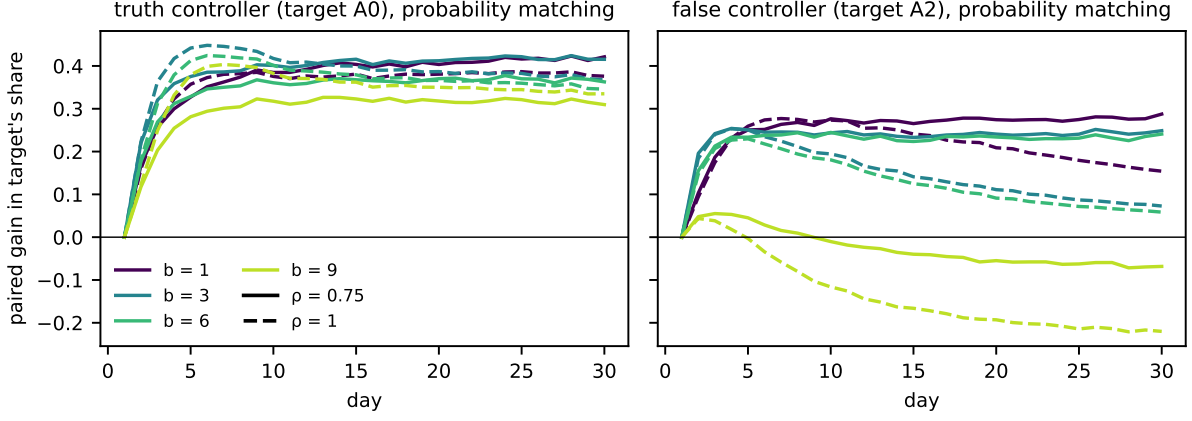


Figure 3: Probability matching: paired gain in the target’s share over the days, by posting budget b ; $q = 6$, $q_c = 12$. Solid $\rho = 0.75$, dashed $\rho = 1$.

- **Both controllers tip the swarm in about 3 days** (Table 1).
- **The truth controller is stronger**, and the gap widens with full memory: gains of about 0.37 against 0.07 at $\rho = 1$. As agents accumulate facts, their pooled evidence pulls back toward the truth’s higher ceiling; the false controller’s gain decays over the days (Figure 3, right, dashed).
- **The reading budget q_c matters little**; the posting budget matters only at its largest value (Section 4).

3 A special case: the A2 drift under argmax voting

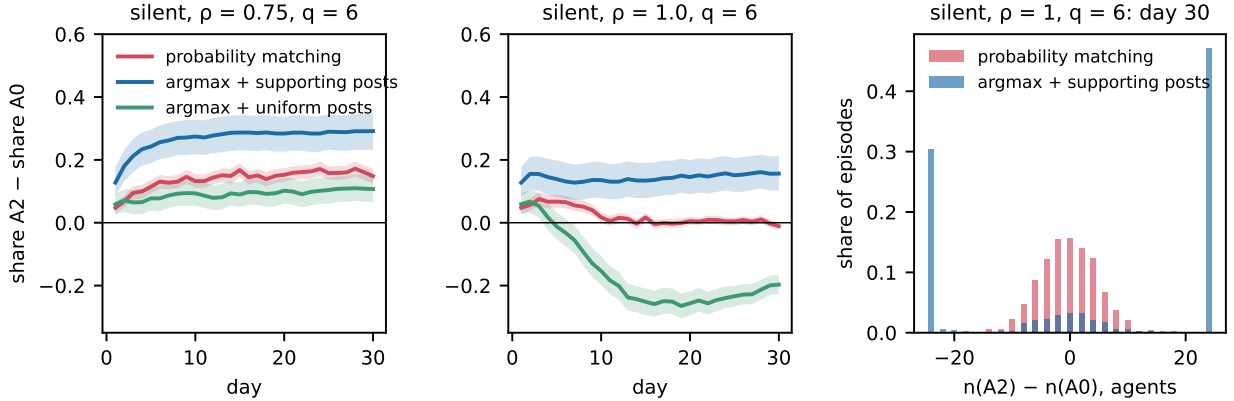


Figure 4: Silent arm, $q = 6$. Left and middle: A2 – A0 vote margin over the days. Right: day-30 spread of the margin across episodes, in agents.

With argmax voting and vote-supporting posts (the *base* rule), the silent swarm does not stay undecided: **63–91% of episodes end unanimous**, all 24 agents on A2 (43–67%) or on A0 (15–41%). The mean margin toward A2 (Figure 4, blue) is the difference between those two groups; the histogram shows the lock-in at ± 24 . Three ingredients combine:

1. **Argmax turns any lead into a vote**, Eq. (2).
2. **Agents post only facts supporting their vote**, so later readers mostly meet the majority’s evidence, and with forgetting the minority’s evidence dies out (day 30, $\rho = 0.75$: A2-side facts held by 53% of agents, A0-side by 24%).
3. **The swarm starts with more A2 voters**. Under argmax, an agent holding a fact that favours two allocations equally votes for each half the time. Its expected starting vote is

$$\mathbb{E}[n_k] = \sum_{j=1}^{24} \frac{\mathbb{1}_{\{k \in \arg \max P_j\}}}{|\arg \max P_j|}, \quad (4)$$

which gives **7 / 6 / 11** for A0 / A1 / A2 (Table 2).

Remove ingredient 1 (probability matching) or 2 (uniform posting: post any remembered fact) and the lock-in collapses: no unanimous episode at all under probability matching, 9–22% under uniform posting.

Table 2: Why the start is 7 / 6 / 11. The setup balanced agents by *fact group* (8 per allocation), assigning a tied fact to the lower-numbered allocation; argmax votes split ties.

	A0	A1	A2
agents per fact group (design bookkeeping)	8	8	8
of which hold a fact tied A1/A2 (counted as A1)	–	6	–
of which hold a fact tied A0/A1 (counted as A0)	2	–	–
expected starting votes, argmax, Eq. (4)	7	6	11
expected starting votes, probability matching	8.3	7.4	8.3

How to fix the start. Balance the agents on *expected votes*, Eq. (4), instead of on fact groups. Under the current rules this is impossible: none of the 10,438 coin-flip agent sets the builder can produce has equal expected A0 and A2 votes. The reason is scarcity: only 3 eligible facts favour A1 alone, so any 5-fact A1 group must include A1/A2-tied facts. Dropping the fact-group rule and requiring expected votes of exactly 8 / 8 / 8 directly works: a random search found such agent sets (still a coin flip, all facts eligible), e.g. 12 facts each held by 2 agents. A full rebuild would also re-check the pool rules (no shared facts, no joint proof). This changes the frozen setup, so it is a decision, not yet applied. Probability matching needs no fix: its start is already balanced.

4 More budget can be worse: a design effect

At the largest posting budget the controller’s gain falls (Figures 1, 2, top row; Figure 3, $b = 9$), and the false controller’s can even turn negative (-0.08 at $\rho = 0.75$, -0.22 at $\rho = 1$, $q_c = 12$, mean over q). The cause is arithmetic, not swarm dynamics. The controller must post exactly b facts or none, and the shared pool holds only n_{target} facts on its target’s side, so each post contains at least

$$c = \max(0, b - n_{\text{target}}) \quad (5)$$

facts that do not favour its target.

Table 3: Pool composition and what the false controller actually posts (probability matching, $\rho = 1$, $q = 6$, $q_c = 12$). Tied facts count for both sides.

	$b = 3$	$b = 6$	$b = 9$
facts in the pool on the target’s side (n_{target})		A0: 4, A2: 6	
posted facts favouring A0 / A1 / A2	14 / 23 / 62%	12 / 22 / 67%	26 / 23 / 51%
nights the controller posts	59%	63%	93%

The fall starts exactly where Eq. (5) becomes positive: at $b = 6$ for the truth controller ($n = 4$), at $b = 9$ for the false one ($n = 6$). Its counter-evidence keeps the target’s vote share below the gate, so it keeps posting (Table 3), and the counter-evidence accumulates. The effect replicates with fresh random draws (a second seed). It disappears if the controller may post *up to* b facts, or never posts facts against its target; in task003-symmetric, whose pools are entirely on their target’s side, more budget simply helps.

5 The controller senses the swarm accurately

Each night the controller estimates the share of agents backing its target from the $n \leq q_c$ agent posts it reads, $\hat{v} = \frac{1}{n} \sum_i \mathbb{1}\{\text{author of post } i \text{ voted for the target}\}$, and stays silent when $\hat{v} \geq 0.75$.

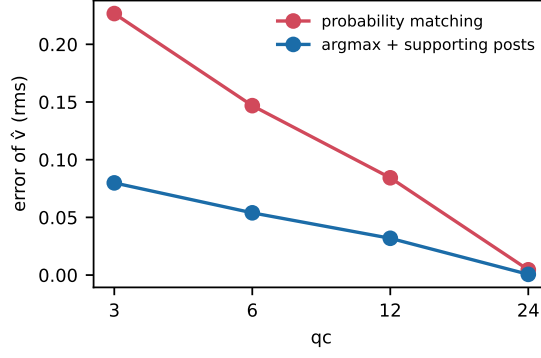


Figure 5: Error of $\hat{v}$ against the true share of all 24 agents, task003-nosolution.

$\hat{v}$ is unbiased (± 0.0005), its error shrinks with q_c and vanishes at $q_c = 24$ (Figure 5). It is larger under probability matching because those swarms stay mixed, which also makes gate mistakes more common: the controller stays silent although the true share is below 0.75 on 5–8% of nights, and posts although it is above 0.75 on up to 13% at $q_c = 3$. Abstaining agents (0.1–0.2% of agent-days) do not bias it.

Helping the truth versus agents’ own proofs: does not apply here. In task003-symmetric, extra truth posts raise the share of agents holding a proof of A0 but lower the share holding one from their own evidence. In task003-nosolution no proof is possible by construction, so this question does not arise.

Caveats. One world (task_003), 24 agents, exact-Bayesian agents; means over 1,000 episodes per cell. Full analysis, code and tables: [analysis/task003_llm_free/sim1_analysis/](#) (reports Q0–Q6).